Computational Physics Lectures

Week 7: Representation, dimensionality reduction, sparsification, and clustering

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Week 7

Representation, dimensionality reduction, sparsification, and clustering

Abstract. Atomic structures must be represented numerically before they can be compared or grouped. This chapter introduces invariant representations and then develops dimensionality reduction through one example with three features and four configurations. The progression is heuristic CUR feature selection, singular value decomposition (SVD), SVD-informed CUR feature selection, and principal component analysis (PCA). Dataset sparsification is discussed separately after dimensionality reduction, followed by distance measures and clustering. Throughout, rows of the data matrix are configurations and columns are descriptor components.

7.1 Introduction

Clustering is an unsupervised-learning task in which unlabeled data are organized into groups. Its outcome depends not only on the clustering algorithm, but also on the representation and distance measure supplied to it.

A flat structure containing N atoms can be specified by $2N$ Cartesian coordinates. This is not an immediately useful machine-learning representation: translation, rotation, reflection, and permutation of identical atoms may alter the coordinate vector without altering the physics. A good representation should be invariant under physically irrelevant transformations and should distinguish physically different structures as far as possible.

The chapter follows the workflow

$$\text{structure} \rightarrow \text{representation} \rightarrow \text{optional dimensionality reduction} \rightarrow \text{distance} \rightarrow \text{clustering}.$$

Dimensionality reduction decreases the number of components used to represent each configuration. We first study heuristic CUR feature selection, then introduce the SVD, use it for informed CUR selection, and finally construct new features by PCA. Dataset sparsification, which instead decreases the number of configurations retained, is discussed only after dimensionality reduction.

7.2 Examples of representations

Pair-distance histogram

An N-atom cluster has $N(N-1)/2$ interatomic distances. A histogram counts how many distances fall in each interval. The descriptor is invariant under translation, rotation, reflection, and atom permutation, but binning loses information.

Smoothed pair-distance distribution

Given pair distances $\{r_i\}$, define

$$g(r) = \sum_i \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(r-r_i)^2}{2\sigma^2}\right].$$

The parameter σ controls a trade-off: a small value preserves detail but makes the representation sensitive, whereas a large value smooths the representation but may hide differences.

If one wishes to evaluate $g(r)$ on a regular histogram grid, $\{x_{\min}, x_{\min}+\Delta x, x_{\min}+2\Delta x, \ldots, x_{\max}\}$, care must be taken with normalization: integrating the continuous Gaussian over each bin width is required to ensure consistency. A consistent discretization is obtained by integrating each Gaussian over the corresponding bin interval:

$$g_j = \sum_i \frac{1}{2}\left[\operatorname{erf}\left(\frac{x_j - r_i + \Delta x/2}{\sqrt{2}\sigma}\right) - \operatorname{erf}\left(\frac{x_j - r_i - \Delta x/2}{\sqrt{2}\sigma}\right)\right]$$

where x_j is the center of the j-th histogram bin. This approach gives the smoothed contribution integrated over bin j. Summing g_j over all sufficiently wide-ranging bins recovers the total number of pair distances.

In bulk systems, the pair distribution function is typically normalized by the expected number of pairs in a shell at radius r, which scales as r in 2D and r^2 in 3D. However, for finite clusters with non-uniform and boundary-dominated structure, such normalization is generally omitted or treated separately.

Connectivity-graph spectrum

Let

$$M_{ij} = \begin{cases} 1, & i \neq j \text{ and atoms } i \text{ and } j \text{ are neighbours,} \\ 0, & \text{otherwise.} \end{cases}$$

The sorted eigenvalues of the real symmetric matrix $\mathbf{M}$ form an N-dimensional descriptor. Relabelling atoms permutes rows and columns but leaves the spectrum unchanged. The representation need not be unique, since different graphs can be cospectral.

Weighted connectivity-graph spectrum

The binary connectivity matrix records only whether two atoms are neighbours. A more general representation assigns a distance-dependent weight to every connected pair:

$$W_{ij} = \begin{cases} f(r_{ij}), & i \neq j, \\ 0, & i = j, \end{cases}$$

where r_{ij} is the distance between atoms i and j. The matrix $\mathbf{W}$ is called a *weighted adjacency matrix* or a *weighted connectivity matrix*, and its sorted eigenvalues form the weighted adjacency spectrum.

The function $f(r)$ determines how strongly atoms at distance r are coupled. For example, one may use a Gaussian weight,

$$f(r) = \exp\left(-\frac{r^2}{2\sigma^2}\right),$$

in which case $\mathbf{W}$ may also be described as a Gaussian kernel, similarity, or affinity matrix. If $f(r)$ is set to zero beyond a cutoff, $\mathbf{W}$ is a sparse weighted adjacency matrix. If $f(r)$ is nonzero for every finite distance, it represents a complete weighted graph.

Coulomb matrix spectrum

This representation resembles the weighted connectivity matrix spectrum, but uses the inverse distance in the off-diagonal elements and the atomic numbers, Z, in the diagonal elements:

$$M_{ij} = \begin{cases} \frac{1}{2} Z_i^{2.4}, & i = j, \\ \dfrac{Z_i Z_j}{r_{ij}}, & i \neq j. \end{cases}$$

Again, the eigenvalues of the matrix are found and sorted in ascending order. The resulting N-dimensional vector is the descriptor for an N-atomic cluster.

7.3 The data matrix

We use one convention throughout:

$$\mathbf{X} = \begin{bmatrix} - - - & \mathbf{x}_1^\mathsf{T} & - - - \\ - - - & \mathbf{x}_2^\mathsf{T} & - - - \\ & \vdots & \\ - - - & \mathbf{x}_N^\mathsf{T} & - - - \end{bmatrix} \in \mathbb{R}^{N \times D}.$$

Each row $\mathbf{x}_n^\mathsf{T} \in \mathbb{R}^{1 \times D}$ represents one configuration, and each column contains one descriptor component across all configurations. Consequently, reducing the number of columns reduces the dimension of the representation, whereas reducing the number of rows reduces the number of configurations in the dataset.

Feature scaling

In the next sections, we shall be introducing centering of features in the course of applying feature dimensionality reduction. Centering removes the mean of each feature, but it does not make features comparable in scale. If descriptor components have different units or substantially different numerical ranges, features with large numerical values may dominate the SVD, PCA, leverage scores, Euclidean distances, and clustering. Depending on the physical meaning of the descriptor, one should therefore consider standardizing the features or applying a physically motivated normalization. Such scaling changes the analysis and should not be introduced automatically without considering the meaning of the features.

7.4 Dimensionality reduction: one example, four viewpoints

We now follow one small example through four stages. The descriptor has $D = 3$ features and the dataset contains $N = 4$ configurations. Our aim is to replace the three-dimensional descriptor by a two-dimensional one, while retaining as much of the relevant variation as possible.

Throughout this section,

$$\mathbf{X} = \begin{bmatrix} -3.00 & -2.75 & 1.00 \\ -1.00 & -1.35 & -1.00 \\ 1.00 & 0.95 & -1.00 \\ 3.00 & 3.15 & 1.00 \end{bmatrix} \in \mathbb{R}^{4\times 3}.$$

Each row is one configuration and each column is one descriptor feature. All three column means are zero, so this particular matrix is already centred. Features 1 and 2 look very similar, whereas feature 3 follows a different variation across the configurations. This observation suggests that two features may be enough.

Step 1: heuristic CUR feature selection

The simplest route to a two-dimensional descriptor is to retain two actual columns of $\mathbf{X}$. This is a CUR-style feature selection: the reduced descriptor consists of original features rather than newly constructed coordinates. The name CUR comes from the standard matrix approximation $\mathbf{X} \approx \mathbf{CUR}$, where $\mathbf{C}$ contains selected columns of $\mathbf{X}$, $\mathbf{R}$ contains selected rows, and $\mathbf{U}$ couples them. Here we use only the column-selection part of this idea because the goal is to select descriptor features.

Before introducing a systematic criterion, we can make a heuristic choice. Feature 2 appears nearly redundant with feature 1, so we retain features 1 and 3 and discard feature 2. With

$$\mathcal{I}_{\text{heur}} = \{1, 3\},$$

the reduced data matrix is

$$\mathbf{X}_{\text{heur}} = \mathbf{X}[:, \mathcal{I}_{\text{heur}}] = \begin{bmatrix} -3 & 1 \\ -1 & -1 \\ 1 & -1 \\ 3 & 1 \end{bmatrix} \in \mathbb{R}^{4\times 2}.$$

All four configurations remain, but each is now described by two actual descriptor components.

Example: justifying the heuristic choice

Because the features are centred, their Pearson correlation coefficient is equivalent to their cosine similarity

$$r_{ij} = \frac{\mathbf{x}_i^\mathsf{T}\mathbf{x}_j}{\|\mathbf{x}_i\|\|\mathbf{x}_j\|},$$

where $\mathbf{x}_i$ denotes column i of $\mathbf{X}$. For this dataset,

$$r_{12} \approx 0.995, \qquad r_{13} = 0, \qquad r_{23} \approx 0.089.$$

Features 1 and 2 therefore vary in almost the same way. Retaining one of them together with feature 3 is a plausible two-feature description. The choice between features 1 and 2 is not yet systematic, however. A heuristic based only on pairwise correlations can also become difficult to apply when many features are present.

Step 2: the singular value decomposition

The heuristic analysis suggested that the three columns of $\mathbf{X}$ do not represent three equally important and independent directions of variation. In particular, features 1 and 2 vary in almost the same way. We would now like a systematic method for identifying the independent linear directions in the data and ranking their importance.

The singular value decomposition (SVD) provides such a method. It rewrites the data matrix in terms of mutually orthogonal directions of variation, ordered according to their importance. This rewriting is an exact factorization of the original matrix. It becomes an approximation only if some of the less important directions are subsequently omitted.

For a matrix

$$\mathbf{X} \in \mathbb{R}^{N\times D}, \qquad f = \min(D, N),$$

the reduced SVD is

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\mathsf{T},$$

where

$$\mathbf{U} \in \mathbb{R}^{N\times f}, \qquad \mathbf{\Sigma} \in \mathbb{R}^{f\times f}, \qquad \mathbf{V} \in \mathbb{R}^{D\times f}.$$

Here "reduced" refers only to the compact dimensions of the matrices in the factorization. The factorization is still exact; dimensionality reduction occurs only if singular-vector terms are subsequently omitted.

The columns of $\mathbf{U}$ and $\mathbf{V}$ are orthonormal,

$$\mathbf{U}^\mathsf{T}\mathbf{U} = \mathbf{I}_f, \qquad \mathbf{V}^\mathsf{T}\mathbf{V} = \mathbf{I}_f,$$

and are called the left and right singular vectors, respectively. The matrix $\mathbf{\Sigma}$ is diagonal,

$$\mathbf{\Sigma} = \operatorname{diag}(\sigma_1, \sigma_2, \ldots, \sigma_f) = \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_f \end{bmatrix},$$

and its nonnegative diagonal entries are the singular values of $\mathbf{X}$. They are ordered as

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_f \geq 0.$$

A singular-vector pair can be understood from

$$\mathbf{X}\mathbf{v}_\alpha = \sigma_\alpha \mathbf{u}_\alpha, \qquad \mathbf{X}^\mathsf{T}\mathbf{u}_\alpha = \sigma_\alpha \mathbf{v}_\alpha.$$

Thus $\mathbf{X}$ maps the unit direction $\mathbf{v}_\alpha$ in feature space onto the unit direction $\mathbf{u}_\alpha$ in configuration space, scaled by σ_α. With configurations stored as rows, $\mathbf{v}_\alpha$ describes how the features combine, while $\mathbf{u}_\alpha$ describes how strongly the configurations participate in the same direction of variation. The singular value σ_α measures the importance of that direction.

Multiplying these relations once more by $\mathbf{X}^\mathsf{T}$ or $\mathbf{X}$ gives

$$\mathbf{X}^\mathsf{T}\mathbf{X}\mathbf{v}_\alpha = \sigma_\alpha^2 \mathbf{v}_\alpha, \qquad \mathbf{X}\mathbf{X}^\mathsf{T}\mathbf{u}_\alpha = \sigma_\alpha^2 \mathbf{u}_\alpha.$$

The right singular vectors are therefore eigenvectors of $\mathbf{X}^\mathsf{T}\mathbf{X}$, and the left singular vectors are eigenvectors of $\mathbf{X}\mathbf{X}^\mathsf{T}$. The numbers σ_α are called *singular values*; the name is historical. Their squares are the common nonzero eigenvalues of $\mathbf{X}^\mathsf{T}\mathbf{X}$ and $\mathbf{X}\mathbf{X}^\mathsf{T}$. Unlike eigenvalues of $\mathbf{X}$ itself, singular values are defined when $\mathbf{X}$ is rectangular.

Connection to covariance

The meaning of covariance, the role of centering, the factor $1/(N-1)$, and the importance of feature scaling are summarized in the appendix *Covariance, variance, and feature scaling*.

If the feature rows have first been centred across the N configurations, the feature covariance matrix is

$$\mathbf{S}_{\text{feature}} = \frac{1}{N-1}\mathbf{X}^\mathsf{T}\mathbf{X} = \mathbf{V}\frac{\mathbf{\Sigma}^2}{N-1}\mathbf{V}^\mathsf{T}.$$

Its eigenvectors are the right singular vectors, and its eigenvalues are $\sigma_\alpha^2/(N-1)$. This is the covariance matrix used in PCA.

The companion matrix

$$\mathbf{G}_{\text{configuration}} = \frac{1}{N-1}\mathbf{X}\mathbf{X}^\mathsf{T} = \mathbf{U}\frac{\mathbf{\Sigma}^2}{N-1}\mathbf{U}^\mathsf{T}$$

is a Gram matrix comparing the centred configurations. It is often called the dual covariance matrix. Its eigenvectors are the left singular vectors. The two matrices have different sizes, $D \times D$ and $N \times N$, so they generally have different total numbers of eigenvalues. However, their nonzero eigenvalues are identical, including their multiplicities. The larger matrix has additional zero eigenvalues.

Example: the common nonzero eigenvalues

For the running example, $\mathbf{X}$ has four rows and three columns. Consequently,

$$\mathbf{X}^\mathsf{T}\mathbf{X} \in \mathbb{R}^{3\times 3}, \qquad \mathbf{X}\mathbf{X}^\mathsf{T} \in \mathbb{R}^{4\times 4}.$$

The two matrices are

$$\mathbf{X}^{\mathsf{T}}\mathbf{X} = \begin{bmatrix} 20.00 & 20.00 & 0.00 \\ 20.00 & 20.21 & 0.80 \\ 0.00 & 0.80 & 4.00 \end{bmatrix}$$

and

$$\mathbf{X}\mathbf{X}^{\mathsf{T}} = \begin{bmatrix} 17.5625 & 5.7125 & -6.6125 & -16.6625 \\ 5.7125 & 3.8225 & -1.2825 & -8.2525 \\ -6.6125 & -1.2825 & 2.9025 & 4.9925 \\ -16.6625 & -8.2525 & 4.9925 & 19.9225 \end{bmatrix}.$$

The eigenvalues of $\mathbf{X}^{\mathsf{T}}\mathbf{X}$ are approximately

$$40.114, \qquad 4.071, \qquad 0.0245,$$

whereas the eigenvalues of $\mathbf{X}\mathbf{X}^{\mathsf{T}}$ are approximately

$$40.114, \qquad 4.071, \qquad 0.0245, \qquad 0.$$

Thus the two matrices have the same three nonzero eigenvalues. The 4×4 matrix has one additional zero eigenvalue because it has one additional dimension.
The common nonzero eigenvalues are the squares of the singular values:

$$\sigma_1^2 \approx 40.114, \qquad \sigma_2^2 \approx 4.071, \qquad \sigma_3^2 \approx 0.0245.$$

Taking the square roots gives the singular values

$$\sigma_1 \approx 6.334, \qquad \sigma_2 \approx 2.018, \qquad \sigma_3 \approx 0.156.$$

Because the data are centred, division by $N - 1 = 3$ gives the eigenvalues of the feature covariance matrix:

$$13.371, \qquad 1.357, \qquad 0.00816.$$

These are the variances that appear later in the PCA calculation.

One may therefore work in feature space or in configuration space, whichever is smaller, without changing the nonzero variance spectrum.

The SVD itself is exact. Writing $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_f]$ and $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_f]$, the factorization can be expanded as

$$\mathbf{X} = \sum_{\alpha=1}^{f} \sigma_\alpha \mathbf{u}_\alpha \mathbf{v}_\alpha^{\mathsf{T}}.$$

Each term is a rank-one contribution associated with one direction of variation. No information is lost when all terms are retained.

Dimensionality reduction begins only when this exact expansion is truncated. Keeping the first $k < f$ terms gives

$$\mathbf{X}_k = \sum_{\alpha=1}^{k} \sigma_\alpha \mathbf{u}_\alpha \mathbf{v}_\alpha^{\mathsf{T}} = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^{\mathsf{T}}.$$

In general,

$$\mathbf{X}_k \approx \mathbf{X},$$

so $\mathbf{X}_k$ is an approximation rather than an exact rewriting of the data. It is the best rank-k approximation to $\mathbf{X}$ in both the Frobenius norm and the spectral norm. The approximation is accurate when the omitted singular values $\sigma_{k+1}, \ldots, \sigma_f$ are small. In particular,

$$\|\mathbf{X} - \mathbf{X}_k\|_F^2 = \sum_{\alpha=k+1}^{f} \sigma_\alpha^2.$$

Example: the SVD reveals two dominant directions

For our 4×3 matrix,

$$\mathbf{V} \approx \begin{bmatrix} 0.705 & 0.111 & 0.700 \\ 0.709 & -0.088 & -0.700 \\ 0.016 & -0.990 & 0.141 \end{bmatrix}, \qquad \mathbf{\Sigma} \approx \begin{bmatrix} 6.334 & 0 & 0 \\ 0 & 2.018 & 0 \\ 0 & 0 & 0.156 \end{bmatrix}.$$

The first right singular vector gives almost equal weight to features 1 and 2, so it describes their common direction of variation. The second is dominated by feature 3. The third is approximately the difference between features 1 and 2, and its small singular value confirms that this difference carries very little information.
The fractions of the squared Frobenius norm carried by the three terms are

$$\frac{\sigma_\alpha^2}{\sum_\beta \sigma_\beta^2} \approx 90.7\%, \qquad 9.2\%, \qquad 0.055\%.$$

Thus the first two terms retain about 99.945% of $\|\mathbf{X}\|_F^2$. The SVD confirms that the data are very nearly two-dimensional, but it has not yet selected which two original features should be retained.

Step 3: informed CUR feature selection

The SVD can be used to replace the heuristic feature choice by an informed one. For a target dimension k, the column leverage score of feature i is

$$\ell_i^{(\text{column})} = \sum_{\alpha=1}^{k} V_{i\alpha}^2.$$

A large leverage score means that the original feature contributes strongly to the dominant k-dimensional right-singular-vector subspace. Because the columns of $\mathbf{V}_k$ are orthonormal,

$$\sum_{i=1}^{D} \ell_i^{(\text{column})} = k.$$

A simple deterministic strategy is to select the columns with the largest leverage scores. Alternatively, columns may be sampled with probabilities

$$p_i = \frac{\ell_i^{(\text{column})}}{k}.$$

For larger problems, leverage scores are often combined with oversampling, rescaling, or a rank-revealing selection method, because selecting the individually largest scores does not by itself guarantee the best-conditioned subset.

Example: leverage scores suggest two original features

For $k = 2$, the row leverage scores are

$$\ell_1 \approx 0.705^2 + 0.111^2 = 0.509,$$
$$\ell_2 \approx 0.709^2 + (-0.088)^2 = 0.510,$$
$$\ell_3 \approx 0.016^2 + (-0.990)^2 = 0.981.$$

Feature 3 has the largest score because it carries the second independent direction of variation almost by itself. Features 1 and 2 have nearly equal scores because they are nearly interchangeable descriptions of the first direction. Selecting the two largest scores gives

$$\mathcal{I}_{\mathrm{CUR}} = \{2, 3\}$$

and hence

$$\mathbf{X}_{\mathrm{CUR}} = \begin{bmatrix} -2.75 & 1.00 \\ -1.35 & -1.00 \\ 0.95 & -1.00 \\ 3.15 & 1.00 \end{bmatrix}.$$

The heuristic choice $\{1, 3\}$ was therefore physically sensible. The SVD-informed calculation refines that choice, but also shows that the difference between selecting feature 1 and feature 2 is marginal.

Informed CUR reduces dimension by *selection*: the resulting axes remain actual descriptor components. This preserves interpretability, but the selected features are not generally the best possible pair of linear coordinates.

Step 4: principal component analysis

Principal component analysis (PCA) takes the alternative approach of constructing new coordinates. It first centres the features,

$$\mathbf{X}_c = \mathbf{X} - \mathbb{1}_N \boldsymbol{\mu}^{\mathsf{T}}, \qquad \boldsymbol{\mu} = \frac{1}{N}\mathbf{X}^{\mathsf{T}}\mathbb{1}_N,$$

where

$$\mathbb{1}_N = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \in \mathbb{R}^N,$$

is the column vector containing N ones. Next, PCA forms the covariance matrix

$$\mathbf{S} = \frac{1}{N-1}\mathbf{X}_c^{\mathsf{T}}\mathbf{X}_c.$$

Diagonalizing

$$\mathbf{S} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{\mathsf{T}}$$

gives principal directions in the columns of $\mathbf{Q}$ and variances in the eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$. The first k principal directions define

$$\mathbf{Z}_k = \mathbf{X}_c\mathbf{Q}_k \in \mathbb{R}^{N\times k}.$$

PCA is directly related to the SVD. If

$$\mathbf{X}_c = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{T}},$$

then

$$\mathbf{S} = \frac{1}{N-1}\mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^{\mathsf{T}}, \qquad \mathbf{Q} = \mathbf{V}, \qquad \lambda_\alpha = \frac{\sigma_\alpha^2}{N-1}.$$

PCA therefore uses the right singular vectors as new descriptor axes.

Example: PCA constructs the two coordinates retaining the most variance

Our example is already centred. Its covariance matrix is

$$\mathbf{S} = \frac{1}{3}\mathbf{X}^{\mathsf{T}}\mathbf{X} \approx \begin{bmatrix} 6.667 & 6.667 & 0.000 \\ 6.667 & 6.737 & 0.267 \\ 0.000 & 0.267 & 1.333 \end{bmatrix}.$$

The eigenvalues are approximately

$$\mathbf{\Lambda} \approx \operatorname{diag}(13.371, 1.357, 0.00816).$$

The first two principal components retain

$$\frac{13.371 + 1.357}{13.371 + 1.357 + 0.00816} \approx 99.945\%$$

of the total variance. Using the first two columns of $\mathbf{V}$ gives

$$\mathbf{Z}_2 = \mathbf{X}\mathbf{V}_2 \approx \begin{bmatrix} -4.049 & -1.080 \\ -1.678 & 0.998 \\ 1.363 & 1.017 \\ 4.364 & -0.935 \end{bmatrix}.$$

The first PCA coordinate is essentially the common motion of features 1 and 2. The second is essentially feature 3. Unlike CUR, PCA does not choose two original features; it constructs the two linear combinations that retain the greatest possible variance.

What the four steps have shown

For this example, all four stages tell a consistent story:

1. Heuristic CUR notices that features 1 and 2 are redundant and keeps one of them together with feature 3.

2. SVD reveals two dominant directions of variation and one very weak difference direction.
3. Informed CUR uses leverage scores to select two actual features, here features 2 and 3.
4. PCA constructs two new coordinates that retain the maximum possible variance.

Thus CUR and PCA solve related but different problems. CUR is preferable when retaining recognizable descriptor components matters. PCA is preferable when the most accurate linear two-dimensional representation matters more than retaining the original axes.

7.5 Dataset sparsification after dimensionality reduction

The preceding discussion reduced the number of descriptor features from $D = 3$ to $D_{\mathrm{red}} = 2$ while retaining all $N = 4$ configurations. A separate task is to reduce the number of configurations. We call this *dataset sparsification*.

With our convention, configurations are rows. Selecting a set $\mathcal{J}$ of configurations gives

$$\mathbf{X}_{\mathrm{sparse}} = \mathbf{X}[\mathcal{J}, :] \in \mathbb{R}^{N_{\mathrm{sel}} \times D}, \qquad N_{\mathrm{sel}} < N.$$

SVD-informed row selection uses the left singular vectors. For a target rank k, the row leverage scores are

$$\ell_j^{(\mathrm{row})} = \sum_{\alpha=1}^{k} U_{j\alpha}^2.$$

This resembles informed CUR feature selection, but its purpose is different: column selection reduces descriptor dimension, whereas row selection reduces the number of configurations.

Sparsification should therefore be considered only after the distinction has been made clear:

$$\begin{array}{ll} \text{dimensionality reduction:} & D \to D_{\mathrm{red}}, \quad N \text{ unchanged}, \\ \text{dataset sparsification:} & N \to N_{\mathrm{sel}}, \quad D \text{ unchanged}. \end{array}$$

They may eventually be combined in a full CUR matrix approximation, but that construction is not needed for the present dimensionality-reduction example.

7.6 Distance measures

Once configurations have vector representations, structural similarity can be quantified by a distance.

Euclidean distance

$$d_{\mathrm{Euc}}(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^{D} (x_i - y_i)^2 \right)^{1/2}.$$

It is sensitive to feature scales, so standardization may be needed.

Cosine distance

$$d_{\cos}(\mathbf{x}, \mathbf{y}) = 1 - \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}.$$

It compares direction rather than magnitude.

7.7 Clustering

k-means clustering

Given k clusters, k-means minimizes

$$J = \sum_{j=1}^{k} \sum_{\mathbf{x}_i \in C_j} \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2.$$

It alternates between assigning every point to its nearest centroid and replacing each centroid by the mean of its assigned points. The result can depend on initialization. PCA is often applied before k-means, but low variance does not necessarily imply irrelevance for clustering.

Steps of the k-means algorithm

1. Initialization:

   The algorithm begins by selecting k initial centroids, denoted $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \ldots, \boldsymbol{\mu}_k$, which represent the centers of the clusters. These centroids can be selected randomly from the dataset or initialized in other ways.

2. Assignment Step:

   Each data point $\mathbf{x}_i$ is assigned to the cluster whose centroid is closest in terms of Euclidean distance. This step minimizes the within-cluster variance for a fixed set of centroids. Mathematically, for each data point $\mathbf{x}_i$, its cluster assignment c_i is given by:

   $$c_i = \underset{j \in \{1,\ldots,k\}}{\arg\min} \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2$$

   where $\|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2$ is the squared Euclidean distance between the data point $\mathbf{x}_i$ and the centroid $\boldsymbol{\mu}_j$. With cluster assignment c_i, a data point belongs to cluster, C_{c_i}.

3. Update Step: After assigning each data point to a cluster, the centroids are updated by calculating the mean of all the points in each cluster. The new centroid $\boldsymbol{\mu}_j$ of cluster j is given by:

   $$\boldsymbol{\mu}_j = \frac{1}{|C_j|} \sum_{\mathbf{x}_i \in C_j} \mathbf{x}_i$$

   where C_j is the set of points assigned to cluster j, and $|C_j|$ is the number of points in that cluster.

4. Convergence:

   The assignment and update steps are repeated iteratively until the cluster assignments no longer change, the decrease in the objective function becomes negligible, or a prescribed maximum number of iterations is reached.

Limitations

- The k-means algorithm is sensitive to the initial selection of centroids, which may lead to suboptimal clustering results. Different initializations can result in different final clusterings. This issue can be mitigated by the **k-means++** initialization method, which selects the first centroid uniformly at random, and each subsequent centroid with probability proportional to the squared distance from the closest already chosen centroid. This strategy spreads out the initial centroids, often leading to faster convergence and improved clustering quality.

- Because k-means minimizes squared Euclidean distances to centroids, it is most effective for compact, roughly convex clusters with comparable spreads and densities. It may perform poorly for elongated, nonconvex, strongly overlapping, or very differently sized clusters.

- k-means can get stuck in local minima, and it is often beneficial to run the algorithm multiple times with different initializations.

- If no points are assigned to a cluster, its mean is undefined. A practical implementation must therefore reinitialize or otherwise handle the corresponding centroid.

Despite these limitations, k-means remains a simple and effective clustering method for many practical applications.

DBSCAN clustering

DBSCAN uses a neighbourhood radius ϵ and a minimum neighbourhood size minPts. A point p is a core point when

$$|N_\epsilon(p)| \geq \text{minPts}, \qquad N_\epsilon(p) = \{q \mid d(p,q) \leq \epsilon\}.$$

With this definition, $p \in N_\epsilon(p)$, so minPts includes the point itself. A point satisfying the criterion is called a core point. A non-core point that lies within the neighbourhood of a core point may be assigned to the same cluster as a border point. Points that are not density-reachable from any core point are labelled as noise.

DBSCAN does not require the number of clusters in advance, can identify nonconvex groups, and labels low-density points as noise. It can be sensitive to its parameters and to varying densities.

Read more about the method at Wikipedia.

7.8 Appendix: Covariance, variance, and feature scaling

Covariance describes how two features vary together across a dataset. For configurations stored as rows, let X_{ni} be the value of feature i for configuration n, and let

$$\mu_i = \frac{1}{N} \sum_{n=1}^{N} X_{ni}$$

be the mean of feature i. The centred values are

$$(X_c)_{ni} = X_{ni} - \mu_i.$$

The sample covariance between features i and j is then

$$S_{ij} = \frac{1}{N-1} \sum_{n=1}^{N} (X_c)_{ni} (X_c)_{nj}.$$

In matrix form,

$$\mathbf{S} = \frac{1}{N-1} \mathbf{X}_c^{\mathsf{T}} \mathbf{X}_c.$$

The diagonal element S_{ii} is the variance of feature i. It measures how strongly that feature varies across the configurations. An off-diagonal element S_{ij} measures whether features i and j tend to vary together. A positive value means that they tend to increase and decrease together, while a negative value means that one tends to increase when the other decreases. A value near zero indicates little linear co-variation, but does not in general imply that the features are independent.

Centering is essential because covariance is intended to describe variation about the feature means. Without centering, $\mathbf{X}^{\mathsf{T}}\mathbf{X}$ also contains contributions from the mean values and is therefore a Gram matrix rather than a covariance matrix.

The factor $1/(N-1)$ is used when the available configurations are regarded as a sample from a larger population. One degree of freedom is lost because the sample mean is estimated from the same data. If the configurations constitute the complete population of interest, one may instead use $1/N$. This choice changes the numerical eigenvalues by an overall factor, but it does not change the eigenvectors or the principal directions.

Covariance depends on the units and numerical scales of the features. If one feature is measured on a much larger scale than another, it can dominate the covariance matrix and hence the PCA directions. This may be appropriate when the scale has physical meaning. Otherwise, one may standardize each feature,

$$\widetilde{X}_{ni} = \frac{X_{ni} - \mu_i}{s_i}, \qquad s_i = \sqrt{S_{ii}},$$

so that every nonconstant feature has unit sample variance. The covariance matrix of the standardized data is the correlation matrix. PCA based on the covariance matrix and PCA based on the correlation matrix therefore answer different questions: the former preserves the original feature scales, whereas the latter gives equal initial weight to all standardized features. The choice should be made from the physical meaning of the descriptor rather than applied automatically.

7.9 Appendix: The Frobenius norm and the spectral norm

Two matrix norms are commonly used when discussing the accuracy of a low-rank approximation: the Frobenius norm and the spectral norm. They measure different aspects of the size of a matrix.

The Frobenius norm

For a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, the Frobenius norm is

$$\|\mathbf{A}\|_F = \left(\sum_{i=1}^{m} \sum_{j=1}^{n} |A_{ij}|^2 \right)^{1/2}.$$

It measures the combined magnitude of all matrix elements. If the singular values of $\mathbf{A}$ are $\sigma_1, \sigma_2, \ldots, \sigma_f$, then

$$\|\mathbf{A}\|_F = \left(\sum_{\alpha=1}^{f} \sigma_\alpha^2 \right)^{1/2}.$$

For the rank-k approximation obtained by truncating the SVD,

$$\|\mathbf{X} - \mathbf{X}_k\|_F^2 = \sum_{\alpha=k+1}^{f} \sigma_\alpha^2.$$

The squared Frobenius norm therefore measures the total squared reconstruction error over all elements of the data matrix.

The spectral norm

The spectral norm of $\mathbf{A}$ is

$$\|\mathbf{A}\|_2 = \max_{\|\mathbf{x}\|_2 = 1} \|\mathbf{A}\mathbf{x}\|_2 = \sigma_1.$$

It measures the largest factor by which the matrix can stretch a unit vector. For the truncated SVD,

$$\|\mathbf{X} - \mathbf{X}_k\|_2 = \sigma_{k+1}.$$

Thus the spectral norm measures the largest reconstruction error in any single direction.

Interpretation for the truncated SVD

The two norms provide complementary interpretations:

- The Frobenius norm measures the total reconstruction error across all matrix elements or, equivalently, across all omitted singular directions.
- The spectral norm measures the worst-case reconstruction error in any one direction.

The truncated SVD is the best approximation of rank at most k under both norms. More precisely, for every matrix $\mathbf{B}$ with $\operatorname{rank}(\mathbf{B}) \leq k$,

$$\|\mathbf{X} - \mathbf{X}_k\|_F \leq \|\mathbf{X} - \mathbf{B}\|_F, \qquad \|\mathbf{X} - \mathbf{X}_k\|_2 \leq \|\mathbf{X} - \mathbf{B}\|_2.$$